Project Design Phase-I
Solution Architecture

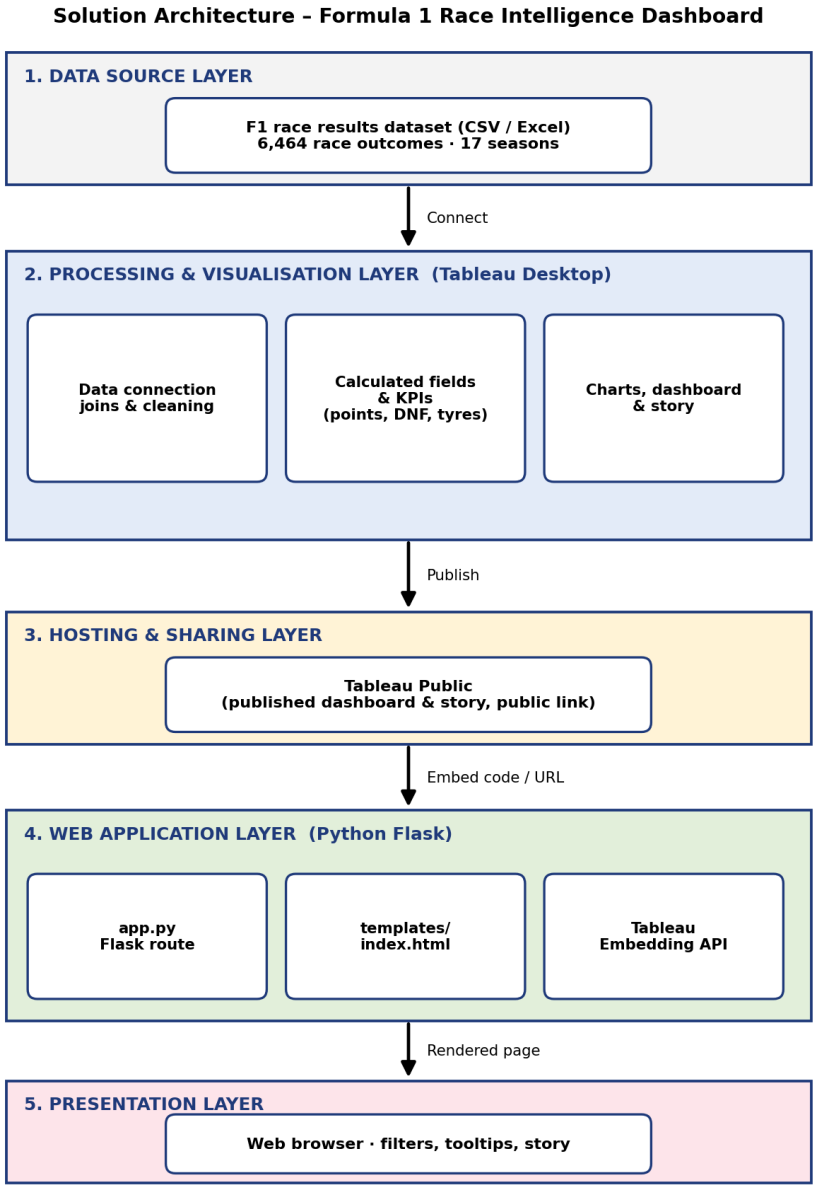
Date	27 October 2026
Team ID	EN25CS3011046
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	F1 race results dataset covering 6,464 race outcomes across 17 seasons, with driver, team, circuit and race details.	CSV / Excel files
Data Preparation & Modelling	Connects the files, joins the tables, cleans values and creates calculated fields such as points, DNF reasons and tyre share.	Tableau Desktop (data connections, joins, calculated fields)
Visualisation & Story	Builds the interactive charts, filters, dashboard and the story from global overview to championship success.	Tableau Desktop
Hosting & Sharing	Publishes the workbook and provides a public link and embed code for the dashboard and story.	Tableau Public
Web Application	A Flask route renders an HTML page that embeds the dashboard and story with Dashboard and Story tabs.	Python, Flask, HTML, CSS
Presentation / End User	Users open the page in a browser, apply filters, hover for tooltips and follow the story.	Web browser, Tableau Embedding API (JavaScript)
Version Control & Documentation	Stores the project code, workbook, templates, video link and documentation for review.	Git, GitHub